VIDYA NIRMAN STUDY NOTES

Class 9 SST: Life on Earth

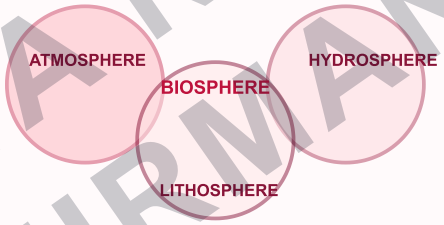
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1. Introduction to the Biosphere

Earth is the only known planet in the Solar System that harbors life. Life exists within a narrow zone of contact where the atmosphere (air), hydrosphere (water), and lithosphere (land) intersect and interact. This narrow life-supporting envelope extending from the deepest ocean trenches to high mountain peaks is called the **Biosphere**.

2. Components of the Biosphere

The biosphere consists of both living organisms and non-living physical elements that continuously exchange matter and energy. It is broadly categorized into two major components: **Abiotic (non-living)** and **Biotic (living)** elements. The seamless interaction between these components maintains global ecological equilibrium.



Key Takeaway

The **Biosphere** is not an independent layer but an active zone where land, air, and water interact to support all forms of plant, animal, and microbial life.

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3. Ecosystems and Ecological Hierarchy

An **ecosystem** is a functional unit of nature where living organisms interact among themselves and with their surrounding physical environment. Ecosystems vary in size from a tiny pond to a vast tropical rainforest or ocean.

Levels of Organization	Definition	Key Examples
Individual Organism	A single living organism capable of independent functioning.	A single Bengal Tiger, a Pine tree.
Population	A group of individuals of the same species in a given area.	All tigers in the Sundarbans forest.
Community	Populations of different species living and interacting together.	Tigers, deer, trees, and fungi in a forest.
Ecosystem	Biotic community interacting with non-living physical factors.	Pond ecosystem, Forest ecosystem.
Biome	Large terrestrial region characterized by distinct climate & flora.	Tropical Rainforest, Tundra, Desert.

Detailed Components of an Ecosystem

A. Abiotic Components

Abiotic factors include climate factors like solar radiation, temperature, rainfall, humidity, and wind, as well as edaphic (soil-related) factors like soil texture, pH, and mineral nutrients. These physical parameters determine the type of species that can survive in a region.

B. Biotic Components

Biotic factors consist of living organisms categorized by their ecological roles: **Producers (Autotrophs)** like green plants, **Consumers (Heterotrophs)** like herbivores and carnivores, and **Decomposers (Saprotrophs)** like bacteria and fungi.

Fact File: Autotrophs vs Heterotrophs

Autotrophs convert solar energy into chemical energy via photosynthesis. **Heterotrophs** depend directly or indirectly on autotrophs for energy, driving the flow of life across ecosystems.

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4. Energy Flow in Ecosystems

Energy is the fundamental requirement for all life processes. Sun is the primary source of energy for all ecosystems on Earth. The flow of energy in an ecosystem is unidirectional—it moves from primary producers to higher trophic levels and is lost as heat at each step.

Trophic Levels

Organisms occupy a specific place in the energy chain according to their feeding relationship, known as their **trophic level**:

Structure of Trophic Levels

- **First Trophic Level (T_1):** Primary Producers (Green plants, phytoplankton).
- **Second Trophic Level (T_2):** Primary Consumers / Herbivores (Deer, grasshoppers, zooplankton).
- **Third Trophic Level (T_3):** Secondary Consumers / Primary Carnivores (Frogs, wolves).
- **Fourth Trophic Level (T_4):** Tertiary Consumers / Apex Predators (Lions, eagles).

5. Food Chains and Food Webs

Energy transfer between organisms occurs through feeding pathways known as food chains and food webs:

A. Food Chain

A linear sequence of organisms through which energy and nutrients pass as one organism eats another. For example: *Grass $\rightarrow$ Grasshopper $\rightarrow$ Frog $\rightarrow$ Snake $\rightarrow$ Eagle.*

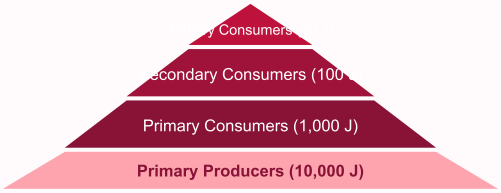
B. Food Web

In nature, simple linear food chains rarely exist independently. A **food web** is a network of interconnected food chains. Food webs provide alternative pathways for energy flow, enhancing the ecological stability of an ecosystem.

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6. Ecological Pyramids & Energy Transfer Rule

Ecological pyramids are graphical representations of ecological parameters (number, biomass, or energy) across successive trophic levels in an ecosystem.



The 10 Percent Energy Law

Lindeman's 10% Rule

Proposed by Raymond Lindeman in 1942, this rule states that during energy transfer from one trophic level to the next, **only about 10% of the energy** is stored as biomass. The remaining 90% is lost as metabolic heat during respiration and life activities.

Types of Ecological Pyramids

1. **Pyramid of Numbers:** Shows the total number of individual organisms at each trophic level (can be upright or inverted).
2. **Pyramid of Biomass:** Represents total organic dry weight at each level.
3. **Pyramid of Energy:** Displays total energy available at each level. *The Pyramid of Energy is ALWAYS upright* because energy decreases at higher levels.

> **Key Takeaway**
>
> Because energy decreases drastically at higher trophic levels, food chains are usually limited to 4 or 5 steps. Top predators require vast land territories to gather sufficient energy.

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7. Biogeochemical Cycles (Nutrient Cycles)

Unlike solar energy which flows unidirectionally out of the Earth system, chemical nutrients (carbon, nitrogen, water, oxygen) are recycled continuously between living organisms and the physical environment through **Biogeochemical Cycles**.

A. The Carbon Cycle

Carbon is the core element of all organic molecules. Plants absorb atmospheric carbon dioxide (CO_2) during photosynthesis to form carbohydrates. Carbon returns to the atmosphere via respiration, decomposition of dead organic matter, and fossil fuel combustion.

B. The Nitrogen Cycle

Nitrogen forms 78% of atmospheric gas but cannot be absorbed directly by most plants. It must undergo transformation:

- **Nitrogen Fixation:** Soil bacteria (e.g., *Rhizobium*) convert N_2 into ammonia (NH_3).
- **Nitrification:** Nitrifying bacteria convert ammonia into nitrites and nitrates.
- **Assimilation & Denitrification:** Plants absorb nitrates; denitrifying bacteria return excess nitrogen to the atmosphere.

8. The Oxygen and Water Cycles

The **Water Cycle (Hydrological Cycle)** involves continuous movement of water through evaporation, transpiration, condensation, and precipitation. The **Oxygen Cycle** is closely linked to photosynthesis (releasing O_2) and respiration (consuming O_2).

Ecological Imbalance from Human Interruptions
Excessive burning of fossil fuels disrupts the Carbon Cycle, increasing atmospheric CO_2 concentration and driving global warming. Overuse of chemical fertilizers disrupts the Nitrogen Cycle, causing water pollution known as *eutrophication*.

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9. Terrestrial Biomes of the World

A **biome** is a large natural regional unit defined by its dominant vegetation, wildlife, and climate patterns. Biomes are primarily determined by average annual temperature and precipitation.

1. Tropical Rainforest Biome

Located near the Equator (e.g., Amazon Basin, Congo Basin, Western Ghats). Characterized by high temperatures and heavy rainfall year-round. Dense, multi-layered canopy vegetation supports over 50% of Earth's plant and animal species.

2. Temperate Deciduous Forest Biome

Found in mid-latitude regions (North America, Europe, East Asia). Experiences four distinct seasons. Trees (e.g., Oak, Maple, Beech) shed their broad leaves in autumn to conserve moisture during cold winters.

3. Grassland Biomes

Grasslands occur in regions with intermediate rainfall insufficient for dense forest growth:

- **Tropical Grasslands (Savanna):** Warm temperatures with wet-dry seasons; tall grasses with scattered trees (e.g., African Savanna).
- **Temperate Grasslands:** Cold winters, hot summers; highly fertile dark soils ideal for agriculture (e.g., Prairies in North America, Steppes in Eurasia, Pampas in South America).

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10. Desert, Taiga, and Tundra Biomes

4. Desert Biome

Covers regions receiving less than 25 cm of rainfall annually. Deserts can be hot (e.g., Sahara, Thar) or cold (e.g., Gobi, Ladakh). Plants are xerophytic—featuring thick fleshy stems, deep roots, and leaves modified into spines to minimize transpiration loss.

5. Taiga (Boreal Coniferous Forest) Biome

Extends across high northern latitudes below the Arctic Circle (Canada, Scandinavia, Russia). Characterized by long, severe winters and short summers. Dominant vegetation consists of evergreen needle-leaf trees (Pine, Spruce, Fir).

6. Tundra Biome

Located in extreme polar regions (Arctic Tundra) and high mountain tops (Alpine Tundra). Characterized by freezing temperatures, brief summers, and a permanently frozen subsoil layer known as **permafrost**.

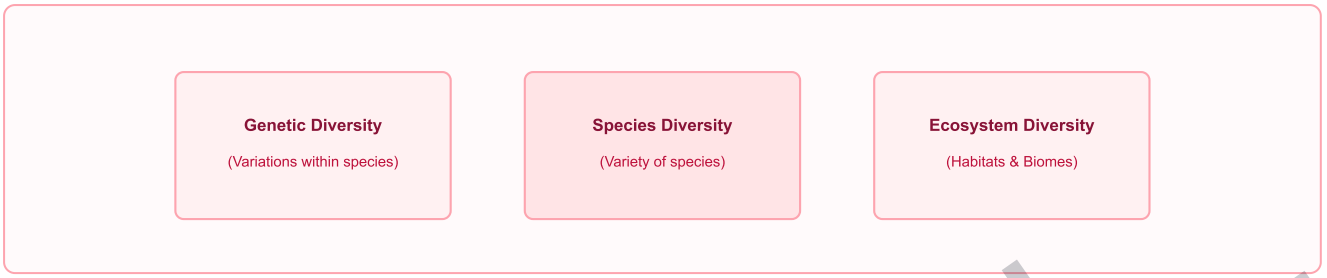
Adaptations in Tundra Life

- **Flora:** Trees cannot grow due to permafrost. Vegetation consists of low-growing mosses, lichens, and dwarf shrubs.
- **Fauna:** Animals possess thick fur, layers of subcutaneous fat (blubber), and white winter coats for camouflage (e.g., Polar Bear, Arctic Fox, Reindeer).

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11. Biodiversity: Concept and Levels

Biodiversity (biological diversity) refers to the variety and variability of life on Earth across all levels of biological organization. Biodiversity underpins ecosystem health, resilience, and human survival.



The Three Levels of Biodiversity

- **Genetic Diversity:** Variations in genetic information among individuals of a single species (e.g., thousands of rice varieties in India).
- **Species Diversity:** The number and abundance of different species present in a defined habitat (e.g., Western Ghats have greater amphibian species diversity than Eastern Ghats).
- **Ecosystem Diversity:** The variety of ecosystems across a geographical region (e.g., India's deserts, rainforests, mangroves, and coral reefs).

Biodiversity Hotspots
Regions featuring exceptionally high species richness and high levels of *endemism* (species found nowhere else) under severe threat of destruction. India hosts four hotspots: Western Ghats, Eastern Himalayas, Indo-Burma region, and Sundaland.

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12. Threats to Biodiversity & Conservation Strategies

Biodiversity is declining globally at an unprecedented rate due to human activities. Understanding these threats is essential for implementing targeted conservation strategies.

Major Causes of Biodiversity Loss (The Evil Quartet)

- **Habitat Destruction & Fragmentation:** Deforestation for agriculture, mining, and urbanization shrinks wild animal habitats.
- **Over-exploitation:** Excessive hunting, poaching, and logging drive species (e.g., Passenger Pigeon) to extinction.
- **Invasive Alien Species:** Non-native species introduced into new habitats displace native flora and fauna (e.g., *Lantana camara*, Water Hyacinth).
- **Co-extinctions:** Extinction of a key species leads to the extinction of dependent organisms (e.g., specialized plant-pollinator pairs).

13. Conservation Approaches

Conservation strategies are categorized into two primary methods:

A. In-situ Conservation (On-Site)

Protecting endangered species within their natural habitat through designated protected networks: **National Parks**, **Wildlife Sanctuaries**, and **Biosphere Reserves** (e.g., Kaziranga National Park, Nilgiri Biosphere Reserve).

B. Ex-situ Conservation (Off-Site)

Protecting threatened animals and plants outside their natural habitats in controlled artificial environments: **Zoological Parks**, **Botanical Gardens**, **Gene Banks**, and **Seed Vaults**.

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14. Quick Revision Summary

- The **Biosphere** is the narrow life-supporting zone intersecting atmosphere, lithosphere, and hydrosphere.
- Energy flows unidirectionally through trophic levels following Lindeman's **10% Energy Law**.
- Nutrients recycle through **Biogeochemical Cycles** (Carbon, Nitrogen, Oxygen, Water).
- Biodiversity exists across 3 levels: Genetic, Species, and Ecosystem. Conservation is carried out **In-situ** or **Ex-situ**.

15. Practice Examination Questions

Section A: Multiple Choice Questions (1 Mark Each)

1. According to the 10% Law of energy transfer, if 10,000 Joules of solar energy is captured by green plants, how much energy is available to primary consumers?
   a) 10,000 J b) 1,000 J c) 100 J d) 10 J
Answer: b) 1,000 J

2. Which biome is characterized by needle-leaf evergreen trees and permafrost subsoil?
   a) Tropical Rainforest b) Temperate Deciduous c) Taiga & Tundra d) Savanna
Answer: c) Taiga & Tundra

Section B: Short Answer Questions (3 Marks Each)

1. Explain the difference between a Food Chain and a Food Web. Why are food webs ecologically more stable?
2. Differentiate between In-situ and Ex-situ biodiversity conservation with suitable examples.

Section C: Long Answer Questions (5 Marks Each)

1. Describe the structure and functioning of an ecosystem, detailing its biotic and abiotic components.
2. What are Biogeochemical Cycles? Explain the Carbon Cycle with reference to human influences driving global warming.

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